Write Strength, Not Write Count: The β Range Dominates Householder Arity in Gated Delta-Rule Attention

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Abstract

Delta-rule linear attention with a channel-wise forget gate can be made more expressive in two ways: by applying more Householder factors per token (*arity*, R), or by widening the range of the write-strength gate β from the strict $(0,1)$ to the reflection range $(0,2)$. Recent work has concentrated on the first. We report a small controlled factorial measurement suggesting the second matters far more. In a fully paired 2×2 design crossing $R \in \{1,2\}$ with β regime, over two non-solvable group word problems (A_5, S_5) and six seed bundles (48 cells, 3.52 GPU-hours), the β widening at fixed $R=1$ is large and significant in every position band where both arms are above chance, while doubling R under strict β is consistently positive but roughly two orders of magnitude smaller — and the β widening is *free*, adding zero parameters, whereas doubling R costs $+40.3\%$. The $\beta \times R$ interaction is positive in five of eight de-nested position bands. Because Kimi K3's KDA layer fixes $\beta = \mathrm{Sigmoid}(\cdot) \in (0,1)$, a determinant argument shows K3-style nodes cannot obtain a negative-determinant transition at any R; whether they would benefit from $R > 1$ in practice is a separate question our scale cannot settle. Two follow-up experiments (72 further cells) close the two confounds we considered most serious. Repeating the design with capacity-matched R=1 controls in *both* regimes (-0.055% parameter match) leaves the effect intact and slightly larger (A_5, positions 129–256: $+57.09 \to +58.88$pp). A learning-rate sweep spanning $33\times$ shows that *no* rate closes the strict arms' deficit at a length inside the training range: they reach stable near-zero training loss and still fall 9–21pp short of the reflection arms, which rules out the leading optimization explanation. We also report results against our own interest. Our pre-registered mechanism — that reflections help because S_5's odd generator demands $\det = -1$ — is *contradicted*, and the fallback explanation we offer is contradicted too. *No* cell survives Holm correction at $n=6$, and the single band that survived before capacity matching does not survive after it. We report the effect, its size, and the reasons it is suggestive rather than established.

1 Introduction

Linear-attention sequence models trade the quadratic cost of softmax attention for a fixed-size recurrent state, and the delta rule [6, 8] has become the dominant way to write into that state: each token performs a rank-one *erase* followed by a rank-one *write*. A well-documented consequence is limited state-tracking ability. Two remedies have been proposed. DeltaProduct [7] increases the number of Householder factors applied per token, a quantity we call the *arity* R. Separately, Grazzi et al. [3] showed that admitting *negative* eigenvalues in the state transition unlocks state tracking that is unreachable otherwise; in a delta-rule node this is achieved by widening the write-strength gate β from the strict range $(0,1)$ to the reflection range $(0,2)$, since $1-\beta$ is the non-unit eigenvalue of each factor.

These two levers are not independent, and the literature has largely pursued the first. That matters practically, because arity is expensive — each extra factor needs its own key and gate projections — while widening β is a one-character change to a sigmoid that adds no parameters at all. It also matters now, because the largest deployed delta-rule attention variant, the Kimi Delta Attention (KDA) layer of Kimi K3 [5], specifies $\beta_t = \mathrm{Sigmoid}(W_\beta x_t) \in (0,1)$ — strictly inside the non-reflective range.

We test this with a fully paired 2×2 factorial crossing arity $R \in \{1,2\}$ with β regime at otherwise identical geometry, on two non-solvable group word problems, with six seed bundles per cell. The design is deliberately narrow: it is built to estimate one interaction term rather than to establish a state of the art. It is also small, and we have tried to be scrupulous about what a 48-cell, $n=6$, 1.4M-parameter probe can and cannot support. Two things in particular govern how our results should be read, and we put them here rather than in a late limitations section.

The metric is a prefix average, and we analyze around it. Our harness scores mean token accuracy over *all* positions $1..L$ of each evaluation sequence. Consequently the five nominal "lengths" are nested, and a model that is correct below $L=40$ and at chance above it still produces a smooth, gracefully-decaying curve that looks like partial extrapolation. For our strict-β arms this is exactly what happens: a zero-parameter model asserting "correct below 40, at chance above" reproduces their entire length curve to within 1.3pp (Table 3). We therefore report *de-nested position bands* as our primary analysis (Section 5.1) and treat the prefix-averaged numbers as secondary. De-nesting strengthens our A_5 result and *eliminates* two of the seven apparently significant S_5 cells, and we report both directions.

Both strict-β arms are at chance beyond $L = 40$. In every de-nested band past position 40, `R1` and `DP2-strict` score within 2× chance (Table 2). This has a consequence we must state plainly, because it defeats an argument we would otherwise have liked to make: the strict-regime arity contrast cannot bound what capacity is worth, since it is measured between two models that are both floored. We therefore removed the confound by experiment rather than bounding it, running capacity-matched R=1 controls in both regimes (Section 7.2); the interaction survives.

Contributions. *(i)* A controlled factorial measurement of arity against β range at matched geometry, fully paired at the seed-bundle level so every arm within a bundle is scored on a byte-identical evaluation bank (Section 4). *(ii)* The finding that in every band where both arms are measurable, the β effect exceeds the arity effect by one to two orders of magnitude at zero parameter cost (Section 5.3), together with a positive $\beta \times R$ interaction in five of eight de-nested bands (Section 5.1). *(iii)* A determinant observation about K3-style nodes, and a correction to a tempting but wrong reading of K3's new bounded-decay parameterization (Section 3). *(iv)* Two negative results against our own hypotheses: the group-parity account of *why* reflections help is contradicted, and so is the headroom account we offered as a replacement (Section 6). *(v)* Two follow-up experiments, 72 further cells, that close the capacity confound by construction and reject the leading optimization account of the remaining one — at the cost of the single Holm-surviving cell, which does not survive capacity matching (Sections 7.1–7.2).

We claim no new operator and no new theory. The gated Householder operator we measure is already implemented in public kernels and published at scale [2]; the determinant argument of Section 2.3 is a special case of a published result [3], and weaker than it. What is new is the controlled measurement. This is an empirical, partly negative result and we frame it as one.

2 Background

2.1 The gated delta rule

Let $S_t \in \mathbb{R}^{d_k \times d_v}$ be the recurrent state, k_t the (L2-normalized) key, v_t the value, β_t the write strength and $\alpha_t \in (0,1)^{d_k}$ a channel-wise forget gate. The gated delta rule is

$$S_t = \left(I - \beta_t k_t k_t^\top\right) \mathrm{Diag}(\alpha_t)\, S_{t-1} + \beta_t k_t v_t^\top, \tag{1}$$

with readout $\tilde{o}_t = S_t^\top q_t$. Equation (1) is exactly the KDA recurrence of Kimi K3, and specializes to gated DeltaNet [9] and, with α scalar per head, to Mamba-2–style gating [1].

2.2 Householder arity

DELTAPRODUCT [7] applies R rank-one factors per token, each with its own key and write strength:

$$S_t = \left[\prod_{i=1}^{R}\left(I - \beta_t^{(i)} k_t^{(i)} k_t^{(i)\top}\right)\right] \mathrm{Diag}(\alpha_t)\, S_{t-1} + \text{writes}. \tag{2}$$

Each factor is a true Householder reflection when $\beta = 2$.

Notation. We write R for the number of factors per token and call it *arity*. DELTAPRODUCT writes n_h, denoting models DELTAPRODUCT$_{n_h}[\cdot]$ with the admissible eigenvalue range in brackets, and does not use the word "arity"; our R is their n_h, and our arms are their $n_h \in \{1,2\}$ (their n_h=1 is DeltaNet). Arity is not free: the R=2 node needs a second set of key and gate projections.

2.3 The determinant fact, and to whom it belongs

The per-token map acting on S_{t-1} in Equation (2) is $A_t = \left[\prod_i (I - \beta_t^{(i)} k_t^{(i)} k_t^{(i)\top})\right] \mathrm{Diag}(\alpha_t)$. For a unit-norm key, $\det(I - \beta k k^\top) = 1 - \beta$, so

$$\det A_t = \left[\prod_{i=1}^{R}\left(1 - \beta_t^{(i)}\right)\right] \cdot \prod_j \alpha_{t,j}. \tag{3}$$

Since $\alpha_{t,j} > 0$ by construction, the *sign* of $\det A_t$ is governed entirely by the β factors, and if every $\beta^{(i)} \in (0,1)$ then $\det A_t > 0$ *for every* R: raising arity cannot recover a negative determinant that the β range has excluded.

What this does *not* predict. We are careful here, because Equation (3) does not by itself predict the interaction we go on to measure. At $R = 2$ with $\beta \in (0,2)$, two negative factors multiply to a *positive* determinant, so R=2 reflection unlocks no determinant sign that R=1 reflection cannot already reach. Read strictly, Equation (3) predicts a main effect of β and *no* $\beta \times R$ interaction at $R = 2$. Our measured interaction is therefore not explained by this argument, and we do not present it as confirmed theory. The determinant argument licenses exactly one claim — that strict β forecloses negative determinants at every arity — and we restrict ourselves to that.

Attribution. Grazzi et al. [3] study $\mathscr{M}_k^n(\Omega)$, the products of exactly k generalized Householder matrices $I - \beta_i v_i v_i^\top$ with $(1 - \beta_i) \in \Omega$, $\|v_i\| = 1$. Their Proposition 1.3 states that any eigenvalue of $N \in \mathscr{M}_k^n((-1,1])$ is either 1 or has modulus below 1, and that *if in addition* $N \in \mathscr{M}_k^n([0,1])$ *and* $k \le 2$, then it lies in $[0,1] \subset \mathbb{R}$. The $k \le 2$ hypothesis is essential — it was added in a published correction, and its failure for $k \ge 3$ is what DELTAPRODUCT exploits. They conclude such transitions are the identity or non-orthogonal, hence neither reflections nor rotations, and that for $k \le 2$ the model cannot learn parity by their Theorem 1. Their Proposition 1.2 is the permutation statement proper: realizing a permutation matrix requires the eigenvalue set $\{-1, 1\}$. Equation (3) is a determinant-level restatement, for the delta rule with normalized keys, of a spectral fact they establish in far greater generality; it also says strictly less, since at $R \ge 3$ a positive determinant no longer implies non-negative real eigenvalues. We claim no theoretical novelty for it whatsoever.

3 The K3 KDA node

Kimi K3 [5] is a mixture-of-experts model with 2.78T total and 104.2B activated parameters, a 1M-token context, and native vision. Each block places three KDA layers followed by one gated MLA layer — a 3:1 ratio *per block* (§2.1) — with one additional MLA layer at the end of the backbone, giving 69 KDA and 24 MLA layers (Table 1, §3.2) and a global ratio of 2.875:1. K3 uses no explicit positional encoding (§3.4) and applies NoPE to all MLA layers, so positional and recency information reaches the model only through KDA's gating and decay (§2.1.2). The KDA recurrence is Equation (1), and its write strength is

$$\beta_t^h = \mathrm{Sigmoid}\big(W_\beta^h x_t\big) \in (0,1), \tag{4}$$

a plain sigmoid, per head and per token, with no scale factor: we searched the report and β is never scaled, doubled, or rescaled anywhere, with the open interval stated in two independent places (§2.1.1). The remaining projections are $q,k = \mathrm{L2Norm}(\mathrm{Swish}(\mathrm{ShortConv}(W_{q/k}x)))$, $v = \mathrm{Swish}(\mathrm{ShortConv}(W_v x))$, and a decay logit $z = W_\alpha^{\uparrow} W_\alpha^{\downarrow} x + b_\alpha^h$ from a low-rank projection plus a per-head bias.

By Equation (3), K3's KDA layers have $\det A_t > 0$ at every token and head, and this would remain true at any R. That is a statement about what strict β forecloses, and it holds independently of our experiment. Whether a K3-style node would *in practice* benefit from $R > 1$ is an empirical question that our 1.4M-parameter probe cannot settle; we return to this in Section 7. We note the report contains no Householder, arity, or DELTAPRODUCT discussion, and no analysis of KDA's state-tracking properties; the question we ask is not one K3 addresses.

A correction worth making explicitly. K3 introduces a *bounded* decay parameterization (§2.1.1, Eq. 5), $g = g_{\min} \cdot \mathrm{Sigmoid}(e^{A_h} z)$ with $g_{\min} = -5$ fixed and A_h a learnable per-head log-scale initialized to 0, so $\alpha = \exp(g) \in (e^{g_{\min}}, 1)$. K3 reports this replaces the unbounded negative-softplus mapping $g = -e^{A_h}\mathrm{Softplus}(z) \in (-\infty, 0)$ used by Kimi Linear [4]. It is tempting to read this floor as what forecloses reflections. It is not. Because $\alpha = \exp(g) > 0$ for any finite g, $\det \mathrm{Diag}(\alpha) > 0$ under *both* parameterizations, so the floor cannot change the sign of $\det A_t$ at all. Determinant sign is governed solely by β. What forecloses odd permutations in K3 is the sigmoid-parameterized β of Equation (4), which Kimi Linear already had.

K3's stated motivation for the floor is numerical, not representational. The reciprocal cumulative decay $1/\Gamma$ used to rescale keys within a chunk can grow without bound and overflow in finite precision; with $g_{\min} = -5$ the cumulative log-decay over a 16-token tile is bounded, so $1/\Gamma$ stays within BF16 range and *all* causal tiles can use dense Tensor Core matrix multiplications, eliminating the position-pair diagonal path (Fig. 3). No modelling or quality justification is offered, and the report contains no architecture ablations at all — so this is an absence of evidence, not evidence of absence. K3 does note the bounded form is related to lower-bounded recurrence gates in prior work; we read that as a relatedness remark appended after the numerical argument, since the paragraph's topic sentence and the figure caption are both purely computational.

One consequence K3 does not discuss cuts against our own framing: the floor $\alpha > e^{-5} \approx 6.7 \times 10^{-3}$ is a genuine modelling constraint, since near-total single-step forgetting was representable under Kimi Linear's unbounded range and is not under K3's. Given that KDA is also K3's only source of positional information, the decay range is load-bearing in a way the numerical framing does not convey. We raise it rather than let it be raised against us.

4 Experimental setup

Arms. We cross arity $R \in \{1,2\}$ with β regime, giving four arms (Table 1). All use the same gated Householder mixer with the same Triton backend, 4000 steps, batch 64, and no feed-forward blocks. The β regime is a single scale on the sigmoid, $\beta = s \cdot \mathrm{Sigmoid}(z)$ with $s \in \{1,2\}$; it adds no parameters, and the two regimes have byte-identical parameter ledgers. Post hoc, the strict arms reach $\beta_{\max} = 1.0000$ and the reflection arms $\beta_{\max} = 2.0000$, with no record violating its bound.

Two follow-up experiments extend this design; both reuse the same bundle seeds, mixer, backend, and step count, and we report them where they bear on a limitation. *(i) Capacity-matched arms* (Section 7.2): `R1-P` and `R1-refl-P` are the R=1 arms with the R=2 parameter delta spent in feed-forward width, solved analytically to $d_{\text{ffn}} = 174$ for 1,399,756 non-embedding parameters (-0.055% against the R=2 arms), holding d_{model}, head count, state dimension and depth fixed — 24 cells. *(ii) A learning-rate sweep* (Section 7.1): the two strict arms at peak rates $\{3\times10^{-4}, 10^{-3}, 3\times10^{-3}, 10^{-2}\}$, two tasks, three bundles — 48 cells. All 72 cells completed; no cell failed.

Tasks. Two group word problems: given a sequence of generators of a finite group, predict the composed element. We use A_5 (order 60, non-solvable, *both* generators even) and S_5 (order 120, non-solvable, transposition generator *odd*). Chance accuracy is $1/60 = 1.67\%$ and $1/120 = 0.83\%$ respectively. The parity difference motivated the prediction Section 6 reports as falsified.

The metric, stated explicitly. Accuracy is the fraction of correct predictions averaged over *all* L positions of 64 evaluation sequences ($N = 64L$ scored tokens; we verified all 240 reported values are integer multiples of $1/64L$). Training words have length 3–40. Because the metric averages positions $1..L$, the nominal lengths are *nested*: $\mathrm{acc}(512)$ includes positions ≤ 40 that are in-distribution. We therefore define *de-nested*

Table 1: The four arms. "Mixer params" excludes both the token embedding and the output head; it is task-invariant, whereas total parameters differ by task because the head does (+39.7% on A_5, +39.1% on S_5 from R=1 to R=2). The β regime is parameter-free. $\bar{\beta}$ is the mean realized β pooled over both tasks and all six bundles.

Arm	β range	R	Mixer params	$\bar{\beta}$
`R1`	$(0,1)$	1	998,092	0.899
`DP2-strict`	$(0,1)$	2	1,400,524	0.779
`R1-refl`	$(0,2)$	1	998,092	1.800
`Reflection`	$(0,2)$	2	1,400,524	1.658

bands

$$\mathrm{band}(L_{i-1},L_i) = \frac{L_i\,\mathrm{acc}(L_i) - L_{i-1}\,\mathrm{acc}(L_{i-1})}{L_i - L_{i-1}}, \qquad (5)$$

the mean accuracy on positions in $(L_{i-1}, L_i]$ alone, and use these as the primary analysis. Only bands past position 40 are extrapolation.

Pairing, and why $n = 6$ helps. Six seed bundles, 1101–1106, each fixing four independent seed streams (initialization, data, task, evaluation), pairwise distinct in every record. Within a bundle all four arms are scored on the same evaluation bank: we confirmed one distinct `eval_bank_sha256` per (task, bundle) shared by all four arms, and twelve mutually distinct hashes across the twelve groups. Every contrast is a within-bundle difference cancelling evaluation-sampling noise identically in all arms. There were 0 train/eval collisions.

Statistics. For each cell we compute per-bundle differences and aggregate those. The interaction is differenced *per bundle before averaging*, $i_b = d_b^{\mathrm{refl}} - d_b^{\mathrm{strict}}$, preserving pairing; computing it from arm means with an unpaired standard error would be wrong. We report mean, $\mathrm{se} = \mathrm{sd}/\sqrt{6}$ (sd with ddof $= 1$), $t = \mathrm{mean}/\mathrm{se}$, and the one-sided 95% lower bound $\mathrm{L95} = \mathrm{mean} - t_{0.95,5}\,\mathrm{se}$, $t_{0.95,5} = 2.015$. We also report Holm-corrected outcomes across the band family, since we test many cells.

Compute and provenance. All 48 cells of the main sweep ran on one NVIDIA A10G, 48/48 completing, for 3.52 GPU-hours (cells 228–305 s). All 48 carry one probe-source revision, 0 records were rejected, 0 non-finite loss steps occurred. The two follow-ups add 72 cells on the same accelerator, 72/72 completing, for 5.16 GPU-hours (cells 228–302 s), under a single later revision, again with 0 rejected records and 0 non-finite steps. Total: 120 cells, 8.67 GPU-hours, no failed cell in either run.

5 Results

5.1 Accuracy, and what the prefix metric hides

Table 2 gives both views. On the prefix-averaged metric the dominant feature is the β contrast: at $L = 128$ on A_5, moving from strict to reflection β at fixed $R = 1$ takes accuracy from 25.74% to 82.83%, while doubling R at fixed strict β takes it from 25.74% to 27.75%.

De-nesting sharpens this and also disciplines it. In every extrapolation band, both strict arms sit within 2× chance — `R1` on A_5 scores 1.24–1.91% against 1.67% chance. The reflection arms are genuinely above chance for longer: `Reflection` on A_5 reaches 96.62% on positions 65–128 and 27.25% on 257–512. Table 3 makes the point quantitatively: a zero-parameter "correct below 40, at chance above" model reproduces both strict arms' entire length curves to within 1.3pp, and fails on the reflection arms by 12–66pp. The honest reading is that *the strict-β arms do not extrapolate past the training range at all*, and the reflection arms partly do.

This also corrects an interpretation we would otherwise have offered. The strict-regime arity effect on the prefix metric decays monotonically with length (A_5: $+3.31, +3.50, +2.01, +0.98, +0.35$pp), which invites a story about extra factors mattering less as the task hardens. It is dilution: the pure prediction $(40/L) \times \mathrm{effect}(40)$ gives S_5 $+2.67, +1.67, +0.83, +0.42, +0.21$ against observed $+2.67, +1.92, +0.82, +0.38, +0.21$. That curve is one number divided by L, and we withdraw any interpretation of its shape.

5.2 The interaction

Table 4 gives the interaction under both metrics. On de-nested bands it is positive at uncorrected L95 in five of eight bands, from +22.14 to +57.09pp. De-nesting *strengthens* A_5 (+13.47 → +30.14 at 65–128; +35.28 → +57.09 at 129–256) and *eliminates* the two long-length S_5 cells, which reverse sign (+8.57 → −0.77; +4.44 → +0.31). Those two cells were significant only because prefix averaging carried the 41–64 signal forward; their per-bundle interaction values correlate at $\rho \geq 0.99$ with the $L = 64$ cell. We do not count them.

We must be equally direct about multiplicity. Across the ten prefix cells, *no* cell survives Holm or Bonferroni correction — Holm stops at the first comparison ($p = 0.00658$ against a threshold of 0.005). This is structural, not unlucky: with $df = 5$, clearing $\alpha = 0.005$ requires $|t| > 4.03$, and no prefix cell reaches it. On the eight de-nested bands, exactly one survives (S_5, 41–64, $t = 4.26$, $p = 0.00401$) — and that one does not survive in the capacity-matched replication of Section 7.2, where it weakens to $t = 3.64$ and no band clears Holm. So the defensible statement is: *no* band-level interaction is established at family-wise $\alpha = 0.05$ in the better-controlled design, and several are positive and large but individually uncorrected. The count-based framing "k of n cells significant" is especially weak here because the cells are not independent (mean within-task cross-length $r = 0.53$, several at 0.99–1.00).

5.3 β dominates arity, and it is free

Table 5 compares the two levers in the bands where both arms are measurable. On A_5 positions 41–64, widening β at fixed $R = 1$ buys +97.98pp ($t = 87.75$) against arity's +3.81pp

Table 2: Mean accuracy (%), $n = 6$ bundles. **Left:** the prefix-averaged metric our harness reports, over positions $1..L$. **Right:** the same data de-nested into position bands via Equation (5), which is what actually measures behaviour at new lengths. Chance is 1.67% (A_5) and 0.83% (S_5); *italic* marks band values within 2× chance. Both strict arms are at chance in every extrapolation band — the prefix metric disguises this as graceful decay.

		Prefix-averaged, positions $1..L$					De-nested bands, positions in $(L_{i-1}, L_i]$				
Arm	Task	40	64	128	256	512	1–40	41–64	65–128	129–256	257–512
`R1`	A_5	79.19	49.96	25.74	13.83	7.82	79.19	*1.24*	*1.53*	*1.91*	*1.81*
`DP2-strict`	A_5	82.49	53.45	27.75	14.81	8.17	82.49	5.05	*2.04*	*1.87*	*1.53*
`R1-refl`	A_5	99.99	99.70	82.83	46.47	23.63	99.99	99.22	65.97	10.11	*0.78*
`Reflection`	A_5	100.00	100.00	98.31	82.73	54.99	100.00	100.00	96.62	67.15	27.25
`R1`	S_5	74.06	46.98	23.65	12.51	6.63	74.06	*1.86*	*0.31*	*1.37*	*0.76*
`DP2-strict`	S_5	76.73	48.90	24.47	12.89	6.84	76.73	*2.53*	*0.04*	*1.31*	*0.79*
`R1-refl`	S_5	97.07	72.86	37.30	19.58	9.99	97.07	32.50	*1.75*	*1.85*	*0.41*
`Reflection`	S_5	98.09	88.45	56.04	28.53	14.64	98.09	72.40	23.62	*1.02*	*0.75*

Table 3: Why we de-nest. Maximum absolute error, over $L \in \{64, 128, 256, 512\}$, of the *zero-free-parameter* prediction $\mathrm{acc}(L) = \frac{40}{L}\mathrm{acc}(40) + \frac{L-40}{L} \cdot \mathrm{chance}$, i.e. "correct below 40, at chance above". The strict arms are fit to within 1.3pp: their apparent length curve carries no extrapolation signal. The reflection arms deviate by 12–66pp, so they genuinely do extrapolate.

Arm	A_5 max err	S_5 max err
`R1`	0.16	0.38
`DP2-strict`	1.27	0.64
`R1-refl`	50.44	11.87
`Reflection`	65.92	26.84

($t = 2.40$), a ratio of 25.7×; on 65–128, +64.44 against +0.52, or 124.7×; on S_5 41–64, +30.64 against +0.67, or 45.5×. Beyond those bands the comparison stops being meaningful because all four arms approach chance, and we report no ratio there. On the prefix metric the corresponding ratios run 6.3× to 44.9× across all ten cells, but we now regard those as contaminated and give them only for continuity.

The cost asymmetry is the part we consider most robust, because it does not depend on the effect size at all: the β widening adds *zero* parameters — both arms have 998,092 mixer parameters and byte-identical ledgers — whereas doubling arity costs +40.3% of mixer parameters (+39.7% / +39.1% of total, by task). Whatever the true effect sizes, one intervention is free and the other is not.

We do not claim arity does nothing. An earlier framing of this work said arity's effect was "not distinguishable from zero". That is true cell-by-cell and misleading overall. The strict-regime arity effect is positive in 10 of 10 prefix cells (one-sided sign test $p = 0.00098$, more significant than any single interaction cell we report), and pooling all 60 per-bundle differences gives +1.61pp, se = 0.61, $t = 2.66$ — though those 60 values are nested and not independent, so the pooled test overstates its own precision. Our design also had little power: the minimum detectable effect at 80% power is 7.4–12.5pp at short lengths, so a real +3.3pp arity effect would have been invisible to us. The accurate statement is that *arity's benefit under strict β is consistently positive, one to two orders of magnitude smaller than the β effect, and below this design's per-cell resolution* — not that it is absent.

6 Two failed explanations

We stated a mechanistic prediction before running the sweep. It failed. The replacement we then offered also fails against the same data. We report both.

The parity prediction, and its falsification. Equation (3) says reflection β unlocks negative determinants, i.e. odd permutations. Our tasks differ exactly in whether odd permutations are demanded: S_5's transposition generator is odd, so composing a word requires elements of determinant -1; in A_5 both generators are even. We predicted a *large* interaction on S_5 and a *small* one on A_5. We observe the opposite in the bands that matter: de-nested, the interaction is +30.14 and +57.09pp on A_5 at 65–128 and 129–256, against −0.77 and +0.31pp on S_5 in the same bands. Only the 41–64 band favours S_5. The reachability-of-reflections story does not explain our result. As Section 2.3 notes, it never should have: Equation (3) predicts no interaction at $R = 2$ in the first place.

The headroom explanation, also contradicted. We then proposed that $\beta > 1$ helps by making each write more expressive regardless of parity, with A_5 (order 60) simply easier than S_5 (order 120) so its reflection arms retain headroom where S_5's have collapsed. This predicts the interaction should track available headroom. It does not. Across the ten prefix cells, the correlation between interaction and headroom (100 − acc of `R1-refl`) is only $r = +0.53$; within A_5 it fits well ($r = +0.93$) but within S_5 there is no relationship ($r = +0.28$, $\rho = +0.10$), and using the `Reflection` arm's de-nested band accuracy as the regressor gives the *wrong sign* ($r = -0.33$). The specific claim we had drafted — that the S_5 reflection arms at 19.58% "have little room left" — is also indefensible: 19.58% against a 0.83% chance floor is ∼80pp

Table 4: The $\beta \times R$ interaction, differenced per bundle before averaging. **Left:** on the prefix-averaged metric. **Right:** on de-nested position bands, our primary analysis. L95 is the one-sided 95% lower bound ($t_{0.95,5} = 2.015$); **bold** marks L95 > 0 uncorrected. Seven of ten prefix cells and five of eight bands are positive at uncorrected L95. Under Holm correction across the family, **zero** of the ten prefix cells and **one** of the eight bands survive (marked †): with $df = 5$, $p < 0.00625$ needs $|t| > 4.03$ and our maximum band t is 4.26. That single survivor does *not* survive once the arms are capacity-matched (Section 7.2, Table 6), where no band clears Holm; this table reports the uncontrolled square. Note S_5 at 129–256 and 257–512 *reverse sign* once de-nested — they were significant only by inheriting the 41–64 signal through prefix averaging.

Prefix-averaged, positions 1..L					De-nested bands				
Task, L	Inter.	se	t	L95	Task, band	Inter.	se	t	L95
A_5, 40	−3.30	4.25	−0.78	−11.87	A_5, 41–64	−3.03	1.53	−1.97	−6.12
A_5, 64	−3.20	3.09	−1.03	−9.43	A_5, 65–128	**+30.14**	9.34	3.23	**+11.32**
A_5, 128	**+13.47**	4.74	2.84	**+3.93**	A_5, 129–256	**+57.09**	16.08	3.55	**+24.69**
A_5, 256	**+35.28**	9.38	3.76	**+16.37**	A_5, 257–512	**+26.74**	12.56	2.13	**+1.43**
A_5, 512	**+31.01**	10.70	2.90	**+9.45**	S_5, 41–64	**+39.23**†	9.21	4.26	**+20.66**
S_5, 40	−1.65	2.42	−0.68	−6.52	S_5, 65–128	**+22.14**	8.80	2.52	**+4.40**
S_5, 64	**+13.68**	4.23	3.23	**+5.15**	S_5, 129–256	−0.77	0.79	−0.98	−2.35
S_5, 128	**+17.91**	6.33	2.83	**+5.15**	S_5, 257–512	+0.31	0.24	1.31	−0.17
S_5, 256	**+8.57**	3.19	2.69	**+2.15**					
S_5, 512	**+4.44**	1.68	2.64	**+1.05**					

Table 5: β range versus arity, on de-nested bands, restricted to the three extrapolation bands where the compared arms are above chance. Both interventions start from the same R=1 strict baseline; Δ is a paired mean difference in points. The β intervention is parameter-free; arity costs +40.3% mixer parameters. Bands past these are omitted because all arms approach chance, making the ratio meaningless.

Task, band	Lever	Δ	t	Ratio
A_5, 41–64	β	+97.98	87.75	
A_5, 41–64	arity	+3.81	2.40	25.7×
A_5, 65–128	β	+64.44	7.55	
A_5, 65–128	arity	+0.52	1.36	124.7×
S_5, 41–64	β	+30.64	6.05	
S_5, 41–64	arity	+0.67	1.02	45.5×

of headroom, more than A_5 at $L = 256$ (46.47%), where the interaction is four times larger.

Where that leaves the mechanism. We have a measured interaction, a falsified pre-registered mechanism, a falsified fallback, and a determinant argument that predicts no interaction at $R = 2$. We therefore offer *no* mechanism. A reader should treat the interaction as an unexplained empirical regularity. The trainability account of Section 7.1 is the leading alternative, and a learning-rate sweep spanning 33× argues against it, though it does not exhaust the optimization hypotheses. Distinguishing the remaining candidates needs a solvable-group control, a difficulty-matched non-solvable pair, and an $R = 3$ arm (where Equation (3) does predict something, since the $k \leq 2$ hypothesis of Grazzi Prop. 1.3 fails there). We ran none of these.

Table 6: The interaction on de-nested bands, before and after capacity matching. **Unmatched** uses `R1/R1-refl` (998,092 non-embedding parameters) as the R=1 cells; **matched** substitutes `R1-P/R1-refl-P`, feed-forward-widened to 1,399,756 against the R=2 arms' 1,400,524 (−0.055%). Both squares share the same six bundles and byte-identical evaluation banks, so the columns are paired. Matching *raises* the A_5 mid-range estimates and leaves the strict-arity effect at ≈ 0; it also costs the one cell that survived Holm in the unmatched square (see Section 7.2). **Bold** marks uncorrected L95 > 0.

		Unmatch.	Capacity-matched			
Task	Band	Inter.	Inter.	se	t	L95
A_5	1–40	−3.30	−2.46	5.90	−0.42	−14.35
A_5	41–64	−3.03	+14.03	11.54	1.22	−9.22
A_5	65–128	+30.14	**+51.43**	14.77	3.48	**+21.67**
A_5	129–256	+57.09	**+58.88**	14.85	3.96	**+28.95**
A_5	257–512	+26.74	**+25.42**	12.53	2.03	**+0.18**
S_5	1–40	−1.65	+5.28	3.50	1.51	−1.78
S_5	41–64	+39.23	**+52.52**	14.43	3.64	**+23.44**
S_5	65–128	+22.14	**+21.77**	9.08	2.40	**+3.47**
S_5	129–256	−0.77	+0.26	1.03	0.25	−1.81
S_5	257–512	+0.31	−0.30	0.35	−0.87	−1.01

7 Limitations

7.1 The trainability confound, which a learning-rate sweep largely closes

This was the objection we considered most serious, and an earlier draft of this paper stated it more strongly than the evidence supports. We correct the statement and then report an experiment that addresses it.

What the loss traces do and do not show. Ten of 48 runs end with a final logged loss above 0.1, and *all ten are strict-β*

Table 7: No learning rate closes the in-distribution gap. Accuracy at $L = 40$, a length *inside* the training range 3–40, for the strict arms at four peak rates ($n = 3$ bundles each), against the reflection arms of the main sweep as the ceiling. "Fit" is tail-mean training loss over the last 200 steps: at the two highest rates the strict arms fit stably and still fall 9–21pp short, so the deficit is not something a better rate repairs by fitting harder. The optimum is interior to the grid for three of four arm/task pairs; `DP2-strict` on A_5 peaks at the top rate, so its optimum is not bracketed.

Task	Arm	Peak LR	Acc. L=40	Fit
A_5	`R1`	$3{\times}10^{-4}$	56.78	0.538
A_5	`R1`	10^{-3}	76.12	0.163
A_5	`R1`	$3{\times}10^{-3}$	83.66	0.093
A_5	`R1`	10^{-2}	83.61	0.082
A_5	`DP2-strict`	$3{\times}10^{-4}$	62.34	0.419
A_5	`DP2-strict`	10^{-3}	86.63	0.063
A_5	`DP2-strict`	$3{\times}10^{-3}$	88.83	0.049
A_5	`DP2-strict`	10^{-2}	91.33	0.031
A_5 *ceiling* (`R1-refl`/`Reflection`)			99.99/100.00	
S_5	`R1`	$3{\times}10^{-4}$	55.44	0.604
S_5	`R1`	10^{-3}	71.07	0.251
S_5	`R1`	$3{\times}10^{-3}$	76.73	0.170
S_5	`R1`	10^{-2}	71.97	0.246
S_5	`DP2-strict`	$3{\times}10^{-4}$	59.06	0.524
S_5	`DP2-strict`	10^{-3}	76.13	0.190
S_5	`DP2-strict`	$3{\times}10^{-3}$	84.02	0.101
S_5	`DP2-strict`	10^{-2}	82.28	0.115
S_5 *ceiling* (`R1-refl`/`Reflection`)			97.07/98.09	

(`R1` 6/12, `DP2-strict` 4/12; both reflection arms 0/12). A perfectly asymmetric stability failure between the arms being compared is the signature of an optimization difference, not only an expressivity one. Within the 24 strict runs, log final loss correlates with accuracy at $r = -0.47$ ($L = 40$).

We previously described those ten as unconverged. That was wrong, and the error was in our instrumentation rather than in the runs. The trace logs one minibatch every 500 steps, so its final entry is a *single batch drawn at step 3999*, where the one-cycle schedule has decayed the rate to 4×10^{-9} and the weights are effectively frozen. All ten reach 10^{-4} or below after warmup; `R1` on S_5 bundle 1106 reads 10^{-4} at step 3000 and 0.82 at step 3999. Those models fit the training set. What the traces do show is *instability*: two runs are above 0.05 at five of seven sampled steps, which is sustained oscillation rather than one unlucky draw. Instability is the correct name for this confound, and it remains perfectly asymmetric across the regimes.

The in-distribution gap. Isolating the confound sharpened it. The strict arms fit to ≈ 0 training loss and still score 68–89% at $L = 40$ — a length *inside* the training range 3–40 — where `R1-refl` scores 99.99% (A_5) and 97.07% (S_5). That is a generalization gap on data drawn from the training distribution, so it is not length extrapolation and the nesting artifact of Section 5.1 does not touch it.

A learning-rate sweep rules out the optimization explanation. We ran the strict arms at four peak rates spanning 33$\times$ ($3{\times}10^{-4}$, 10^{-3}, $3{\times}10^{-3}$, 10^{-2}), two tasks, three bundles, 48 cells (Table 7). *No rate closes the gap.* The best strict cell anywhere is `DP2-strict` on A_5 at 91.33%, still 8.67pp short of the reflection ceiling; on S_5 the best is 84.02%, 14.06pp short. At the two highest rates the strict runs are not merely fitting but fitting *stably* — tail-median training loss 0.0006–0.0084 over the last 200 steps — and still cap at 86.8%. They fit the training set and do not generalize, and no rate repairs that by fitting harder.

Two caveats keep this from being conclusive. On three of the four arm/task pairs the optimum is interior to the grid, so the grid brackets it; but `DP2-strict` on A_5 peaks at the top rate we tried, so we have not bracketed its optimum and cannot exclude a still-better rate above 10^{-2}. And a learning-rate sweep is not the only optimization lever: schedule shape, weight decay, gradient clipping, and initialization scale are untested. What we can say is that the single most likely optimization explanation, tested over a 33$\times$ range, does not account for a 9–21pp deficit.

We therefore no longer claim the expressivity and trainability accounts are inseparable in general. Over the swept range they are separable, and the evidence favours expressivity. We do continue to flag that $\beta = 2$ doubles the effective write-gate gradient scale, so the reflection arms' perfect stability record may itself be an optimization-geometry effect.

7.2 Parameter matching: the capacity confound, now closed

The R=1 arms have 998,092 mixer parameters and the R=2 arms 1,400,524, +40.3%. Any raw "R=2 beats R=1" main effect is confounded with capacity, and in the four-arm design above we claimed none.

We had intended to argue that the strict-regime arity effect ($\leq +3.50$pp) upper-bounds what that capacity is worth. We withdraw that argument. As Table 2 shows, both strict arms are at chance in every extrapolation band, and a capacity contrast measured between two floored models is guaranteed to read ≈ 0 regardless of what capacity is worth. The bound is uninformative rather than reassuring, and using it would have been circular.

Rather than argue the confound away, we removed it. We ran two additional arms, `R1-P` and `R1-refl-P`, which are R=1 with the R=2 parameter delta spent entirely in feed-forward width at identical d_{model}, head count, state dimension and depth — so the recurrent state size, the variable the comparison must hold fixed, is unchanged. Solving the width analytically gives $d_{\text{ffn}} = 174$ and 1,399,756 non-embedding parameters against `DP2-strict`'s 1,400,524, a -0.055% match, ten times inside our 0.5% tolerance. Both regimes are matched, not just the strict one: the estimand is an interaction, so controlling one contrast and not the other would leave the two halves differently constructed and their difference would mix the correction with the effect. The 24 cells reuse the bundle seeds of the main sweep, and we verified that every new cell's evaluation bank hashes identically to the corresponding cell of the original run, so the matched square is paired with the original arms rather than merely comparable to them.

The interaction survives capacity matching, and strengthens slightly. Table 6 gives both squares side by side. On A_5 at positions 129–256 the de-nested interaction moves from +57.09pp (unmatched) to +58.88pp (matched, se 14.85, $t = 3.96$, L95 +28.95); at 65–128 it moves from +30.14 to +51.43pp. The strict-regime arity effect stays ≈ 0 in every extrapolation band under matching (−0.23 to +3.10pp), which is now an informative statement rather than a circular one, because `R1-P` is no longer capacity-starved relative to `DP2-strict`.

Matching costs us one significance claim, and we report it. Under Holm at $\alpha = 0.05$ over ten de-nested bands, the unmatched square had exactly one survivor (S_5, 41–64, $t = 4.26$, $p = 0.00401$). In the matched square that cell weakens to $t = 3.64$ ($p = 0.00745$) and *no* band survives: the strongest is A_5 129–256 at $p = 0.00535$ against a first-step threshold of 0.00500, which it misses. So capacity matching raises the point estimates on A_5 and lowers the family-wise evidence to zero surviving cells. Both facts belong in the record.

7.3 $n = 6$, multiplicity, and influential bundles

Six bundles makes paired contrasts informative but imprecise: interaction standard errors run 1.68–16.08pp, and the `Reflection` arm's between-bundle spread on A_5 is large (sd 20.19pp at $L = 256$). As Section 5.1 states, no prefix cell and only one band survives Holm correction, and with $df = 5$ none could have cleared Bonferroni even in principle. Leave-one-bundle-out shows the two long-length S_5 prefix cells lose significance if either of two bundles is dropped — consistent with their sign reversal under de-nesting. The A_5 result is more robust: dropping the most extreme bundle (1103) takes $L = 256$ from +35.28 to +29.18 (L95 +10.56) and $L = 512$ from +31.01 to +22.26 (L95 +6.18), both still positive. We report L95 values, not point estimates, as the defensible claim.

7.4 Ceiling and floor censoring

Of 240 reported values, 26 are exactly 100.00% and 35 exceed 99%; of the de-nested band values, 107 of 240 lie within 2× chance and 10 band-cells fall strictly below it. At $L = 40$ the reflection arms score 99.99% and 100.00% on A_5, the latter with zero variance across all six bundles, so the $L = 40$ interaction is mechanically bounded and its negative sign (−3.30pp) is not interpretable. As a robustness check we recomputed the interaction on Haldane–Anscombe-corrected log-odds: sign and significance broadly survive (9 of 10 prefix cells positive at uncorrected L95), but the largest interaction moves from $L = 256$ to $L = 128$ on A_5, and S_5 at $L = 40$ becomes significantly *positive*. So the raw-scale magnitudes are partly an artifact of pp differences being maximized near 50%, and our reading of the $L = 40$ cells as uninformative is right on A_5 but wrong on S_5.

7.5 Scale, scope, and the K3 claim

Our models have 1.0–1.4M mixer parameters, no feed-forward blocks, and train for 4000 steps on synthetic group word problems. Kimi K3 has 2.78T total parameters. **Nothing in this paper is direct evidence about the behaviour of a frontier-scale model.** We separate the two K3 statements accordingly. The determinant observation — that strict β forecloses negative-determinant transitions at every arity — follows from Equation (3) alone, is regime-agnostic, and does not depend on this experiment; we consider it sound. The empirical prediction that K3-style nodes would not *benefit* from $R > 1$ does depend on this experiment, and given the caveats above we present it as a hypothesis motivating a scaled replication, not as a finding. Group word problems are also a deliberately adversarial probe chosen because they discriminate; language modelling may weight these mechanisms differently.

We also flag a tension with the prior work we rely on. DeltaProduct reports that its models "fail to learn even the training context-length when restricted to the standard eigenvalue range $[0, 1]$, regardless of the number of Householder transformations n_h". Our strict-β arms do learn the training range (74–82% at $L = 40$ at our default rate, and up to 91.3% once the rate is tuned, Table 7), while reaching near-zero training loss throughout. So the disagreement is not that their models fail to fit and ours fit: ours fit and then generalize imperfectly. Either the tasks differ materially or our strict arms are not the regime they describe; we cannot say which.

7.6 Other limitations

We measure $R \in \{1, 2\}$ only, so we cannot speak to $R \geq 3$ — where, as noted, the theory actually predicts something different. All arms share one mixer implementation and Triton backend, so a kernel bug interacting with the β scale would not be caught by our design. The main 2×2 sweep uses a single learning-rate schedule per arm; Section 7.1 sweeps the rate for the strict arms only, on three bundles rather than six, so the reflection arms remain untuned and we cannot rule out that a different rate would move *them* as well — though they are already at 97–100% on the metric in question, so there is little room for it to matter.

8 Related work

Delta-rule linear attention. The delta rule as a fast-weight update goes back to Schlag et al. [6]; DeltaNet [8] made it trainable at scale with a chunkwise-parallel form, and gated DeltaNet [9] added the channel-wise forget gate that makes Equation (1) the modern default. Mamba-2 [1] supplies the structured-state-space view of the same gating. Kimi Linear [4] introduced KDA, and Kimi K3 [5] deploys it as the majority attention layer at frontier scale.

Arity and negative eigenvalues are one research programme. The two lines of work our paper sits between share most of their authors. Grazzi et al. [3] established that admitting negative eigenvalues unlocks state tracking otherwise out of reach — the result our Section 2.3 specializes and which subsumes our determinant observation. DELTAPRODUCT [7] then raised arity, and its §5.2 reports the finding that most directly prefigures ours: "Unexpectedly, S_4 and A_5 can extrapolate robustly using only $n_h = 2$ despite the theorem suggesting 3 and 4, respectively." Two details matter for us. They attribute the efficiency to S_4 and A_5 being isomorphic to subgroups of $SO(3,\mathbb{R})$, and they observe a head that learns $\beta = 2$ on *both* factors — two true reflections composing to a rotation. And their §5.2 uses the extended eigenvalue range throughout, reporting the failure under $[0,1]$ quoted in Section 7. That sentence is the closest prior statement to our result. Our contribution is to convert it from a training-time observation reported in passing into a paired factorial estimate at length extrapolation — with the caveats above. Our result is a refinement of their programme, not a challenge to it.

The operator is not new, and we are precise about what is prior. The per-channel-gated delta-rule operator we measure is computed by public kernels for generalized (diagonal-plus-low-rank) delta-rule updates, and is established at scale: EDA [2] trains dense 2.5B and MoE 25B-A2.8B models on it — though it appears there as the *baseline*, explicitly credited to KDA, EDA's own contribution being a second, independently addressed erase factor. To avoid overclaiming on DELTAPRODUCT's behalf: its Appendix B.4 Eq. 6 does place a per-channel $\mathrm{diag}(w)$ inside a product at arbitrary n_h, but the factors are asymmetric RWKV-7 matrices rather than Householders, the construction is offered explicitly as buying expressivity "at the cost of stability", and it is used for a theory result with no experiments; DELTAPRODUCT's own gated variant uses a *scalar* gate outside the product. We contribute no operator, no kernel and no theorem — only a controlled measurement.

9 Conclusion

In a gated delta-rule node, the write-strength range β and the Householder arity R are not interchangeable routes to expressivity. Across 48 paired cells on two non-solvable group word problems, widening β produced large gains in every position band where the compared arms were above chance — 25.7× to 124.7× the gain from doubling arity — at zero parameter cost, against arity's +40.3%. The $\beta \times R$ interaction was positive in five of eight de-nested bands. The determinant argument of Section 2.3, which is not ours, establishes independently of any experiment that a strict-β node such as Kimi K3's KDA layer cannot obtain a negative-determinant transition at any arity.

Two of the confounds that stood between this measurement and a result are now closed, and we closed them by running the experiments rather than by argument. Capacity matching — R=1 controls widened to within 0.055% of the R=2 arms, in *both* regimes — leaves the interaction intact and slightly larger (+57.09 → +58.88pp on A_5 at positions 129–256), which also retires the uninformative bound we had tried to place on it. A learning-rate sweep spanning 33× shows no rate closes the strict arms' deficit at a length *inside* the training range: they reach stable near-zero training loss and still fall 9–21pp short. That is the leading optimization explanation tested and rejected, and it reframes the deficit as a generalization gap rather than a training failure.

What remains open we state plainly. We have no mechanism: our pre-registered explanation is falsified and so is the fallback. *No* band-level interaction survives multiple-comparison correction at $n = 6$ — and the single band that survived before capacity matching does not survive after it, so the better-controlled design has weaker family-wise evidence, not stronger. One arm of the rate sweep is unbracketed at the top of the grid, and optimization levers other than the rate are untested. The remaining fixes are specified and cheap: more bundles, an $R = 3$ arm, and a wider optimization search. What we can defend is narrower than where we started but firmer than in our first draft: one lever is free and appears to matter a great deal, the other is expensive and appears to matter little, the difference is not explained by capacity or by the learning rate, and we do not yet know why or whether it survives at scale.

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